

HANDOUT A

Problem: The goal is not simply to make the benchmark smaller, but it is to find the smallest subset that still preserves model rankings, go/no-go decisions, and difficulty coverage. All-pass and all-fail items carry little discriminative signal but removing them entirely distorts the difficulty distribution. The subset must be both difficulty-calibrated and content-diverse.

Algorithm: Spectral-IRT Stratified Pruner

The algorithm operates in three stages. *Stratify*: a binary ($n_{\text{models}} \times n_{\text{items}}$) score matrix is built from evalscope reviews, items are partitioned by pass-rate into Easy, Medium, and Hard, and the pruning budget is allocated proportionally to preserve the full difficulty distribution. *Cluster*: items within each stratum are embedded using *all-MiniLM-L6-v2* (384-dim) and spectral clustering is run with a Gaussian kernel, with bandwidth σ set via median pairwise distance and the number of clusters equal to the stratum budget. Spectral clustering operates on pairwise similarities, allowing it to handle non-linear structure that k-means would miss. *Select*: one item per cluster is selected: in the Medium stratum, the item with the highest IRT information score [$a^2 \times P \times (1-P)$, peaking at $P = 0.5$] is chosen as it best separates models at the ability boundary, while in Easy and Hard strata the centroid-closest item is chosen since typicality is more useful where models are unanimous.

For MMMU's single-model constraint, two modifications apply. Since binary scores produce no meaningful pass-rate strata, MMMU's *topic_difficulty* labels are used to define the Easy/Medium/Hard partition directly. Since IRT discrimination requires inter-model variance, all strata use centrality selection. The feature matrix also adds a subject one-hot and 6-dim image-type multi-hot vector (Diagrams, Charts, Medical, Photography, Geographic, Tables), each scaled by $\log_2(n_{\text{classes}})$, ensuring all six types are represented. Each type probes a distinct encoder failure mode: charts require quantitative spatial reading, medical images fine-grained texture discrimination, diagrams relational parsing, and photographs broad categorical recognition.

Stratification prevents distribution shift: without it, clustering on the full pool oversamples Easy items and inflates all model scores. Spectral clustering eliminates content redundancy by forcing exactly one item per embedding-space region, and IRT selection concentrates the discriminative budget precisely where models disagree. The result is LCB pruned from 315 to 72 items (4.4 $\times$), AA-LCR from 100 to 24 items (4.2 $\times$), and MMMU from 360 to 42 items (8.6 $\times$). On LCB and AA-LCR, Spearman $\rho = 1.0$ and 100% go/no-go agreement are preserved, and LOMO errors (3.4pp, 2.8pp) are well below overall MAE, confirming generalization beyond the three shipped models. Target sizes are derived from a stratum coverage floor of $\lceil v_{\text{stratum_size}} \rceil$: $n = 48$ was tested on LCB and produced a ranking flip ($\rho = 0.866$), confirming that 72 is the principled minimum. On MMMU, MAE is 0.44pp and KL divergence is effectively zero (4×10^{-5}).

Assumptions: Pass rates are stationary and the three-stratum partition is exact with 3 binary-scored models since only four pass-rate values are possible; IRT estimates are noisy at this scale and the coverage floor is a heuristic, not a theorem. Models are treated as deterministic with stable rankings, the go/no-go threshold requires recalibration for MMMU if a different model is evaluated, and MMMU's difficulty labels are treated as ground truth so miscalibration propagates silently.

What would change with more resources?

More data: compression improves since the coverage floor scales as $\lceil v_{\text{stratum_size}} \rceil$, meaning a 1000-item benchmark would need ~ 100 items rather than 229. Larger pools also stabilize IRT estimates, making a proper 2PL model feasible. *Live endpoint:* The pruner becomes adaptive. A seed set can be scored first, per-item informativeness can be estimated from early responses, and evaluation can be concentrated on boundary items near the go/no-go threshold. This is computerized adaptive testing, where IRT stopping rules terminate evaluation once the ability estimate converges. *More time:* Target size can be cross-validated using model-held-out folds instead of a sensitivity sweep. Embeddings can be fine-tuned on domain-specific text, competitive programming for LCB and vision-language descriptions for MMMU, to produce a better-calibrated similarity space. Clustering can be run over multiple random seeds to eliminate the sensitivity to a single initialization that caused the LCB ranking flip at $n = 48$.